

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B. Pharm. Examination
University Theory Examination November 2016
PH205 Pharmaceutical Engineering-III

Date: 29.11.2016, Tuesday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

- Q 1** Attempt any **TWO** of the following
- A (a) Define Mixing. Describe various factors affecting mixing **05**
(b) Classify various equipments used for mixing. Discuss in detail Planetary Mixer with suitable diagram. **05**
- B Define Crystallization. Briefly describe theory of crystallization. **10**
- C Discuss the Mier's super saturation theory of crystallization. What are the limitations of the Mier's theory? **10**
- Q 2** Attempt any **TWO** of the following
- A Discuss in detail about the elements of automatic process control system. **10**
- B Classify the various flow meters. Write a detail note on orifice meter with suitable diagram. **10**
- C Classify the mixtures. Briefly describe mechanisms of mixing in liquid. **10**

SECTION- II

- Q 3** Attempt any **TWO** of the following
- A Write application of size reduction. Explain principle, working, advantages and disadvantages of Ball mill with suitable diagram **10**
- B (a) Describe with a clean and labelled diagram the construction, principle, working, advantages, disadvantages of cyclone separator. **5**
(b) Write a note on standards sieves. **5**
- C (a) Classify different hazards which occurs in pharmaceutical industry. Explain electrical hazards in detail **5**
(b) Draw diagram of a mill which is useful for size reduction of thermolabile material **5**
- Q 4** Attempt any **TWO** of the following
- A Classify various approaches of particle engineering. Explain any two approaches in detail **10**
- B (a) Describe any one method which is useful for solid- liquid particle separation. **5**
(b) Describe principle and working of following with suitable diagram **5**
 - Hammer mill
 - Triple roller mill
- C Write a note on about effluent treatment for the waste water **10**
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